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nf-cavalier: A Nextflow Pipeline for Rare Disease Variant Prioritization and Reporting

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Summary

nf-cavalier is a Nextflow pipeline that automates genomic variant annotation, filtering, and reporting for individuals with rare Mendelian diseases. It accepts individual, family, or cohort variant callsets together with a target gene panel or phenotype. Variants are filtered by predicted gene consequence, computational pathogenicity, population frequency, and familial segregation, and then visualised for human review. Results are presented in user-friendly interactive HTML reports and PowerPoint slide decks, with links to external resources that enable rapid review by clinical research teams. **nf-cavalier** is maintained on GitHub (bahlolab/nf-cavalier) under the permissive MIT licence.

Statement of need

Identifying candidate pathogenic genetic variants from genome sequencing data is a complex task that requires coordinating a wide range of software and data sources. Millions of variant calls are generated per individual and must be interpreted alongside known disease gene associations (e.g., PanelApp (Martin et al., 2019), HPO (Köhler et al., 2021)), gene and transcript models (e.g., GENCODE (Frankish et al., 2021)), computational pathogenicity predictions (e.g., CADD (Schubach et al., 2024)), population frequency data (e.g., gnomAD (Chen et al., 2024)) and known pathogenic variant databases (e.g., ClinVar (Landrum & Kattman, 2018)). Variant inheritance patterns must be considered in familial cases, with population frequency filters tailored to dominant, recessive, and compound heterozygous inheritance patterns. Finally, candidate variants must be reported, and the sequencing data supporting the variant visualised for human review prior to further validation, segregation, and pathogenicity classification.

This task is further complicated by the need to jointly interpret short variants (single-nucleotide variants (SNVs) and indels (< 50 bp)) and structural variants (larger alterations; > 50 bp), which are typically identified using separate workflows.

Several software tools are commonly used for the individual tasks carried out in this workflow. For example, VEP (McLaren et al., 2016) handles variant annotations and consequence prediction based on specified gene models, but does not support inheritance-based filtering or variant visualisation. slivar (Pedersen et al., 2021) enables rapid querying of variant call format (VCF) files with an expressive syntax that allows specific inheritance models to be interrogated; however, it does not perform annotation or visualisation. Tools such as igv-reports (Robinson et al., 2026) and SVPV (Munro et al., 2017) generate visualisations of sequencing data to aid

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variant review, but do not perform prioritization. This fragmentation highlights the need for a variant prioritization framework that can orchestrate a range of task-specific tools, ultimately improving reproducibility and usability.

nf-cavalier addresses this need by integrating annotation, inheritance and gene-list based filtering, visualisation, and reporting in a single workflow.

State of the field

Comparable rare disease variant prioritization tools include seqr (Pais et al., 2022), Scout (Stranneheim et al., 2021), and MOLGENIS-VIP (Maassen et al., 2025) (Table 1). VIP is the closest comparator to **nf-cavalier**, as both are Nextflow-based and produce static HTML reports without requiring a server or database backend, suiting institutional HPC environments where persistent services may be unsupported. In contrast, seqr and Scout require continuously running web services, creating ongoing infrastructure and maintenance requirements. The recommended seqr deployment uses Kubernetes to orchestrate several databases, the seqr application, and a variant-loading service; Scout uses Docker services for MongoDB and its web application. Many institutional HPC environments restrict Docker and Kubernetes deployments.

Table 1: Comparison of **nf-cavalier** to alternative variant prioritization tools.

Tool	nf-cavalier	MOLGENIS-VIP	seqr	Scout
Publication year	2026	2025	2022	2021
Required Preprocessing	SV and SNV/Indel variant calling	None	GATK4 SNV/Indel calling	nf-core/raredisease variant calling
Dependencies	Nextflow and container engine	Nextflow and container engine	Kubernetes, Live Server	Docker, Live Server
Variant types	SNV/Indel, SV	SNV/Indel, SV, STR	SNV/indel, SV†	SNV/indel, SV, STR
QC	Somalier -relate and -ancestry, Sample contamination	None	None	None
Gene lists supported	PanelApp, HPO, genomic region, HGNC locus, Genes4Epilepsy§, custom	HPO‡	PanelApp, HPO, genomic region, custom	PanelApp, HPO, custom
Filtering	Nextflow parameters, R-script	JSON Decision tree, HTML GUI	Web GUI	Web GUI
Outputs	Static HTML, PowerPoint slides	Static HTML	Web server	Web server

† SV support in seqr is currently limited to in-house users at the Broad Institute, as the SV calling and loading pipelines have not been made available at the time of writing. ‡ VIP does not filter output based on gene panels, but can annotate HPO terms for user-level filtering. § (Oliver et al., 2023)

nf-cavalier and VIP serve different use cases. VIP targets production-oriented clinical

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variant classification, encoding ACMG-derived evidence criteria as a configurable JSON decision tree. **nf-cavalier** instead supports exploratory, tunable research prioritization without automated classification, identifying candidates for downstream review and interpretation. Filtering criteria such as allele frequency, genotype quality, consequence class, and gene-list membership are exposed as independently configurable, named parameters. For example, `FILTER_SHORT_POP_DOM_MAX_AF = 0.0001` sets the maximum gnomAD allele frequency for dominantly inherited short variants. This transparently represents the filtering strategy, allowing researchers to inspect and iteratively refine individual criteria.

Software design

Variant prioritization pipelines are subject to substantial churn – new annotation sources become available, new pathogenicity-prediction methods are released, and new classes of variation are recognised. **nf-cavalier** is designed to incorporate new resources with minimal effort. Its use of the VCF specification provides compatibility with diverse upstream callers and annotation resources, making it particularly suitable for research pipelines at the cutting edge of variant discovery.

The Nextflow implementation supports reproducible, containerised execution on local machines, HPC clusters, and cloud environments. The pipeline consists of three modules: annotation, QC, and reporting (Figure 1). Also included is a one-time setup pipeline that downloads the required reference data, and a script to download a test dataset based on the 1000 Genomes CEPH trio.

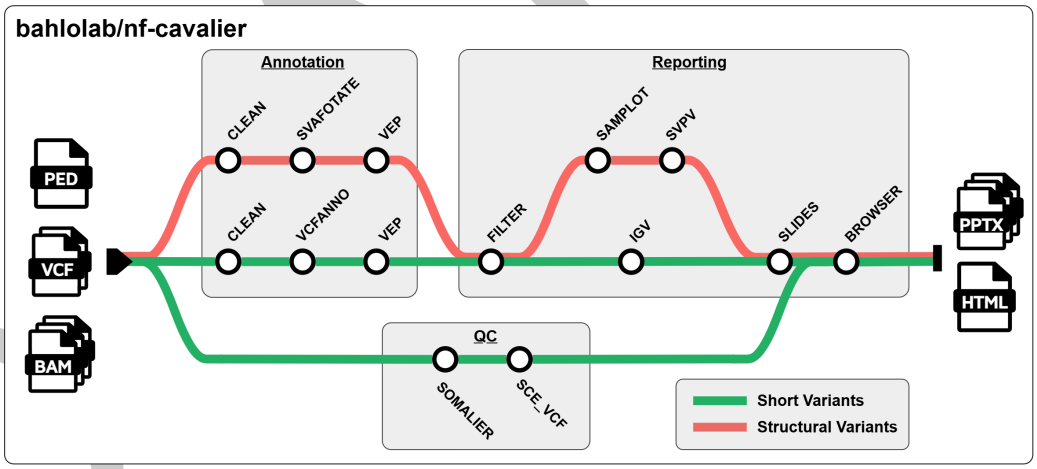


Figure 1: Schematic overview of **nf-cavalier** pipeline.

Annotation takes cohort VCFs (short and/or structural variants) as input, splits them into shards for parallel processing, and performs variant normalisation and annotation with standard tools bcftools (Danecek et al., 2021), vcfanno (Pedersen et al., 2016), VEP (McLaren et al., 2016), and SVAFotate (Nicholas et al., 2022). Sharding improves scalability, reducing runtime from days to hours depending on available compute. vcfanno is employed to rapidly incorporate pre-calculated variant-specific metrics for short variants, including population allele frequencies from gnomAD v4.1, known pathogenicity classifications from ClinVar, predicted deleteriousness scores from CADD, as well as arbitrary user-specified annotation sources. VEP is used to add gene-level consequence predictions as well as annotations that depend on transcript models, including REVEL (Ioannidis et al., 2016), AlphaMissense (Cheng et al., 2023), and SpliceAI (Jaganathan et al., 2019) scores. For structural variants, SVAFotate is used to transfer gnomAD v4.1 SV frequencies using an overlap-based matching approach.

Filtering operates on a per-family basis, applying user-configurable filters for genes of interest (derived from PanelApp, HPO, genomic regions, or custom inputs), genic consequence from VEP, and deleteriousness predictions. By default, all ClinVar pathogenic variants in a gene of interest are retained, and all ClinVar benign variants are excluded. Variants are then matched to inheritance models – all affected family members must carry dominant variants; recessive variants must be present only in a homozygous state in affected family members; and compound heterozygous pairs must be present together only in affected family members. Allele frequency filters are then applied depending on inheritance model, with the default dominant threshold set at 1/10,000 and the default recessive threshold set at 1/100.

The reporting stage visualises candidate variants using igv-reports, SVPV, or Samplot to support manual review and quality control. Reports are compiled into PowerPoint slide decks (Figure 2) and published alongside a static HTML report that enables interactive exploration of variants (Figure 3). The HTML report also provides access to other resources via public APIs, including PanelApp, HPO, ClinVar, gnomAD, and the UCSC genome browser.

In addition to the annotation, filtering, and reporting stages, the pipeline includes a QC workflow that runs somalier (Pedersen et al., 2020) and SCE-VCF (Lu et al., 2023) to assess relatedness, sex concordance, genetic ancestry, and contamination, enabling users to identify problematic samples and verify pedigree relationships.

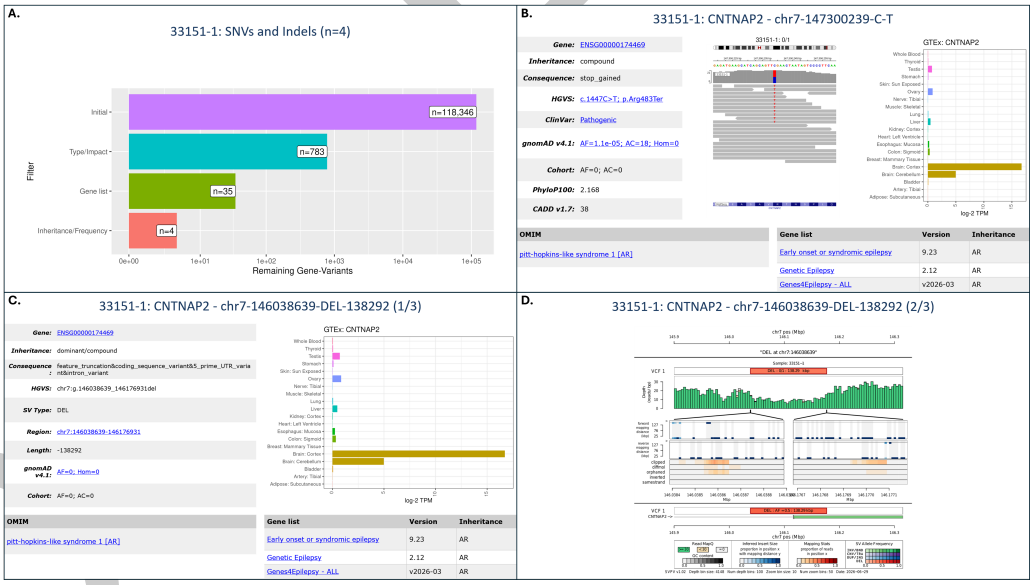


Figure 2: Example **nf-cavalier** variant PowerPoint output slides for a research participant with compound heterozygous pathogenic variants in *CNTNAP2*. A) Summary slide detailing counts after each level of SNV/Indel filtering was applied, resulting in four candidate variants for this example analysis. B) PowerPoint output slide with details on a stop-gain variant known to be pathogenic in ClinVar. Blue text represents hyperlinks to external resources. C) PowerPoint output slide detailing a 138 kb deletion affecting *CNTNAP2*. D) nf-cavalier PowerPoint slide embedding an SVPV plot of the structural variant for inspection and QC. This case has been published in Munro et al. (Munro et al., 2026).

Research impact statement

nf-cavalier has been used in several studies investigating known and novel causes of rare disease in large-scale genome and exome sequencing cohorts. These include the Austin Health Adult Undiagnosed Disease Program (AHA-UDP), the first adult-focused undiagnosed disease program in Australia, which resulted in diagnoses for 16/50 (32%) participating families (Wallis et al., 2024). More recently, **nf-cavalier** has been used to search for pathogenic variants in a

large cohort of individuals with developmental and epileptic encephalopathy (DEE), resulting in diagnoses for 37/242 (15%) of participants, including nine structural variants (Munro et al., 2026), and in the identification of a complex structural variant in FBRSL1 associated with a severe DEE (Cohen-Vig et al., 2026). Furthermore, **nf-cavalier** was used to assist in the landmark discovery of an intronic repeat expansion in FGF14 as the most common cause of late-onset cerebellar ataxia, where it was used to exclude alternative explanations for pathogenicity (Rafehi et al., 2023). The CAVALIER R package, a precursor to the current Nextflow implementation, was also used similarly to assist with the discovery of the RFC1 repeat expansion causing cerebellar ataxia with neuropathy and bilateral vestibular areflexia syndrome (CANVAS) (Rafehi et al., 2019).

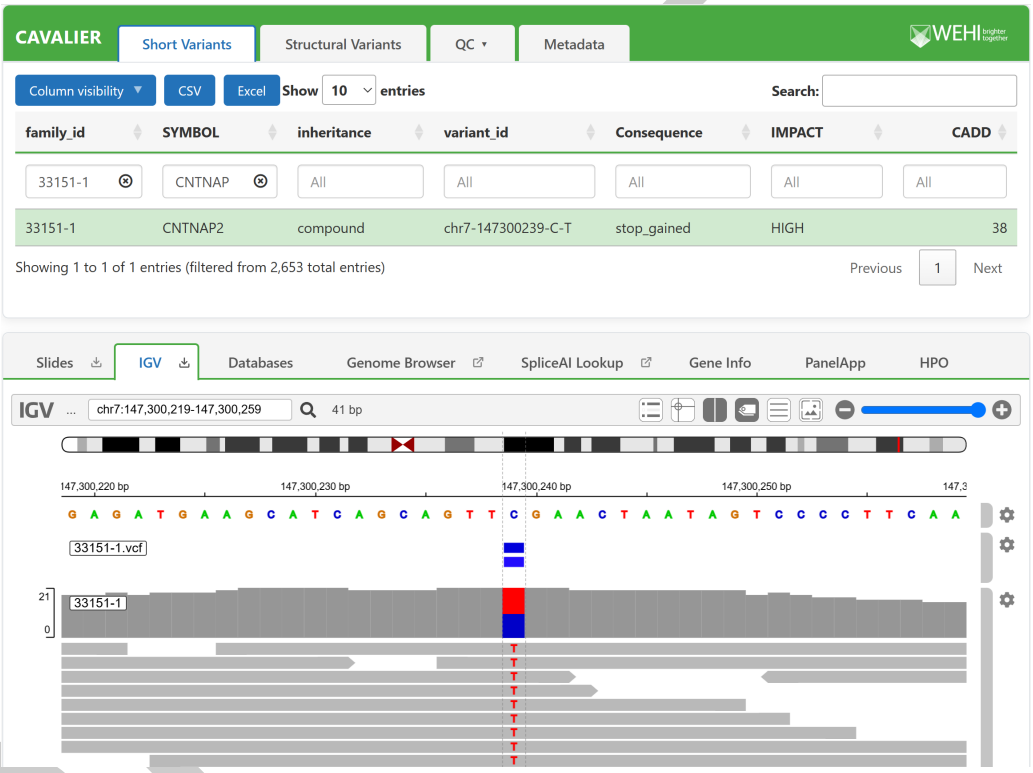


Figure 3: The CAVALIER interactive variant browser interface. The upper panel allows filtering of variants and includes tabs for QC data and run metadata. The lower panel enables exploration of variants and genes, with igv-reports (pictured) and other external APIs embedded.

AI usage disclosure

The majority of the **nf-cavalier** source code was written and tested without generative AI assistance. Since March 5, 2026 (GitHub commit 'a1196d9'), Claude Code has been used for coding and documentation assistance, using a mixture of Sonnet 4.6 and Opus 4.7-4.8 models. AI assistance was most useful for creating the interactive HTML output, expanding the documentation, and implementing the pipeline to download the reference data. The AI tools were given clear, specific instructions on implementation and design, and all AI-generated content has been reviewed and tested by the authors. The authors made key decisions and take full responsibility for the software's reliability and maintenance. The Claude web interface has also been used to assist with grammar, clarity, and phrasing for this manuscript.

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References

Chen, S., Francioli, L. C., Goodrich, J. K., Collins, R. L., Kanai, M., Wang, Q., Alföldi, J., Watts, N. A., Vittal, C., Gauthier, L. D., Poterba, T., Wilson, M. W., Tarasova, Y., Phu, W., Grant, R., Yohannes, M. T., Koenig, Z., Farjoun, Y., Banks, E., … Karczewski, K. J. (2024). A genomic mutational constraint map using variation in 76,156 human genomes. *Nature*, *625*(7993), 92–100. https://doi.org/10.1038/s41586-023-06045-0

Cheng, J., Novati, G., Pan, J., Bycroft, C., Žemgulytė, A., Applebaum, T., Pritzel, A., Wong, L. H., Zielinski, M., Sargeant, T., Schneider, R. G., Senior, A. W., Jumper, J., Hassabis, D., Kohli, P., & Avsec, Ž. (2023). Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science*, *381*(6664), eadg7492. https://doi.org/10.1126/science.adg7492

Cohen-Vig, L., Munro, J. E., Reid, J., Witkowski, T., Sikta, N., Kraus, D., Bennett, M. F., Scheffer, I. E., Hildebrand, M. S., Bahlo, M., & Berkovic, S. F. (2026). A complex duplication overlapping FBRSL1 implicated in a developmental and epileptic encephalopathy. *medRxiv*. https://doi.org/10.64898/2026.03.30.26349353

Danecek, P., Bonfield, J. K., Liddle, J., Marshall, J., Ohan, V., Pollard, M. O., Whitwham, A., Keane, T., McCarthy, S. A., Davies, R. M., & Li, H. (2021). Twelve years of SAMtools and BCFtools. *Gigascience*, *10*(2). https://doi.org/10.1093/gigascience/giab008

Frankish, A., Diekhans, M., Jungreis, I., Lagarde, J., Loveland, J. E., Mudge, J. M., Sisu, C., Wright, J. C., Armstrong, J., Barnes, I., Berry, A., Bignell, A., Boix, C., Carbonell Sala, S., Cunningham, F., Di Domenico, T., Donaldson, S., Fiddes, I. T., García Girón, C., … Flicek, P. (2021). GENCODE 2021. *Nucleic Acids Res.*, *49*(D1), D916–D923. https://doi.org/10.1093/nar/gkaa1087

Ioannidis, N. M., Rothstein, J. H., Pejaver, V., Middha, S., McDonnell, S. K., Baheti, S., Musolf, A., Li, Q., Holzinger, E., Karyadi, D., Cannon-Albright, L. A., Teerlink, C. C., Stanford, J. L., Isaacs, W. B., Xu, J., Cooney, K. A., Lange, E. M., Schleutker, J., Carpten, J. D., … Sieh, W. (2016). REVEL: An ensemble method for predicting the pathogenicity of rare missense variants. *Am. J. Hum. Genet.*, *99*(4), 877–885. https://doi.org/10.1016/j.ajhg.2016.08.016

Jaganathan, K., Kyriazopoulou Panagiotopoulou, S., McRae, J. F., Darbandi, S. F., Knowles, D., Li, Y. I., Kosmicki, J. A., Arbelaez, J., Cui, W., Schwartz, G. B., Chow, E. D., Kanterakis, E., Gao, H., Kia, A., Batzoglou, S., Sanders, S. J., & Farh, K. K.-H. (2019). Predicting splicing from primary sequence with deep learning. *Cell*, *176*(3), 535–548.e24. https://doi.org/10.1016/j.cell.2018.12.015

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Köhler, S., Gargano, M., Matentzoglu, N., Carmody, L. C., Lewis-Smith, D., Vasilevsky, N. A., Danis, D., Balagura, G., Baynam, G., Brower, A. M., Callahan, T. J., Chute, C. G., Est, J. L., Galer, P. D., Ganesan, S., Griese, M., Haimel, M., Pazmandi, J., Hanauer, M., … Robinson, P. N. (2021). The human phenotype ontology in 2021. *Nucleic Acids Res.*, *49*(D1), D1207–D1217. https://doi.org/10.1093/nar/gkaa1043

Landrum, M. J., & Kattman, B. L. (2018). ClinVar at five years: Delivering on the promise. *Hum. Mutat.*, *39*(11), 1623–1630. https://doi.org/10.1002/humu.23641

Lu, W., Gauthier, L. D., Poterba, T., Giacopuzzi, E., Goodrich, J. K., Stevens, C. R., King, D., Daly, M. J., Neale, B. M., & Karczewski, K. J. (2023). CHARR efficiently estimates contamination from DNA sequencing data. *Am. J. Hum. Genet.*, *110*(12), 2068–2076. https://doi.org/10.1016/j.ajhg.2023.10.011

Maassen, W. T. K., Johansson, L. F., Charbon, B., Hendriksen, D., Hoek, S. van den, Slofstra, M. K., Mulder, R., Meems-Veldhuis, M. T., Sietsma, R., Lemmink, H. H., Diemen, C. C. van, Gijn, M. E. van, Swertz, M. A., & Velde, K. J. van der. (2025). MOLGENIS VIP: An end-to-end DNA variant interpretation pipeline for research and diagnostics configurable to support rapid implementation of new methods. *NAR Genom. Bioinform.*, *7*(2), lqaf087. https://doi.org/10.1093/nargab/lqaf087

Martin, A. R., Williams, E., Foulger, R. E., Leigh, S., Daugherty, L. C., Niblock, O., Leong, I. U. S., Smith, K. R., Gerasimenko, O., Haraldsdottir, E., Thomas, E., Scott, R. H., Baple, E., Tucci, A., Brittain, H., Burca, A. de, Ibañez, K., Kasperaviciute, D., Smedley, D., … McDonagh, E. M. (2019). PanelApp crowdsources expert knowledge to establish consensus diagnostic gene panels. *Nat. Genet.*, *51*(11), 1560–1565. https://doi.org/10.1038/s41588-019-0528-2

McLaren, W., Gil, L., Hunt, S. E., Riat, H. S., Ritchie, G. R. S., Thormann, A., Flicek, P., & Cunningham, F. (2016). The ensembl variant effect predictor. *Genome Biol.*, *17*(1), 122. https://doi.org/10.1186/s13059-016-0974-4

Munro, J. E., Dunwoodie, S. L., & Giannoulatou, E. (2017). SVPV: A structural variant prediction viewer for paired-end sequencing datasets. *Bioinformatics*, *33*(13), 2032–2033. https://doi.org/10.1093/bioinformatics/btx117

Munro, J. E., Thiyagarajah, H., Bennett, M. F., Chiu, A. T. G., Schneider, A. L., Bennett, C. A., Lieffering, N., Allan, T., Witkowski, T., Harris, R. V., Reid, J., Sikta, N., Macdonald, S., Coulter, L., Dang, Y. L., Kerkhof, J., Sadikovic, B., Perucca, P., Berkovic, S. F., … Hildebrand, M. S. (2026). Genome sequencing boosts diagnostic yield for the developmental and epileptic encephalopathies. *medRxiv*, 2026.04.24.26351703. https://doi.org/10.64898/2026.04.24.26351703

Nicholas, T. J., Cormier, M. J., & Quinlan, A. R. (2022). Annotation of structural variants with reported allele frequencies and related metrics from multiple datasets using SVAFotate. *BMC Bioinformatics*, *23*(1), 490. https://doi.org/10.1186/s12859-022-05008-y

Oliver, K. L., Scheffer, I. E., Bennett, M. F., Grinton, B. E., Bahlo, M., & Berkovic, S. F. (2023). Genes4Epilepsy: An epilepsy gene resource. *Epilepsia*, *64*(5), 1368–1375. https://doi.org/10.1111/epi.17547

Pais, L. S., Snow, H., Weisburd, B., Zhang, S., Baxter, S. M., DiTroia, S., O'Heir, E., England, E., Chao, K. R., Lemire, G., Osei-Owusu, I., VanNoy, G. E., Wilson, M., Nguyen, K., Arachchi, H., Phu, W., Solomonson, M., Mano, S., O'Leary, M., … O'Donnell-Luria, A. (2022). Seqr: A web-based analysis and collaboration tool for rare disease genomics. *Hum. Mutat.*, *43*(6), 698–707. https://doi.org/10.1002/humu.24366

Pedersen, B. S., Bhetariya, P. J., Brown, J., Kravitz, S. N., Marth, G., Jensen, R. L., Bronner, M. P., Underhill, H. R., & Quinlan, A. R. (2020). Somalier: Rapid relatedness estimation for cancer and germline studies using efficient genome sketches. *Genome Med.*, *12*(1), 62.

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https://doi.org/10.1186/s13073-020-00761-2

Pedersen, B. S., Brown, J. M., Dashnow, H., Wallace, A. D., Velinder, M., Tristani-Firouzi, M., Schiffman, J. D., Tvrdik, T., Mao, R., Best, D. H., Bayrak-Toydemir, P., & Quinlan, A. R. (2021). Effective variant filtering and expected candidate variant yield in studies of rare human disease. *NPJ Genom Med*, *6*(1), 60. https://doi.org/10.1038/s41525-021-00227-3

Pedersen, B. S., Layer, R. M., & Quinlan, A. R. (2016). Vcfanno: Fast, flexible annotation of genetic variants. *Genome Biol.*, *17*(1), 118. https://doi.org/10.1186/s13059-016-0973-5

Rafehi, H., Read, J., Szmulewicz, D. J., Davies, K. C., Snell, P., Fearnley, L. G., Scott, L., Thomsen, M., Gillies, G., Pope, K., Bennett, M. F., Munro, J. E., Ngo, K. J., Chen, L., Wallis, M. J., Butler, E. G., Kumar, K. R., Wu, K. H., Tomlinson, S. E., … Lockhart, P. J. (2023). An intronic GAA repeat expansion in FGF14 causes the autosomal-dominant adult-onset ataxia SCA50/ATX-FGF14. *Am. J. Hum. Genet.*, *110*(1), 105–119. https://doi.org/10.1016/j.ajhg.2022.11.015

Rafehi, H., Szmulewicz, D. J., Bennett, M. F., Sobreira, N. L. M., Pope, K., Smith, K. R., Gillies, G., Diakumis, P., Dolzhenko, E., Eberle, M. A., Barcina, M. G., Breen, D. P., Chancellor, A. M., Cremer, P. D., Delatycki, M. B., Fogel, B. L., Hackett, A., Halmagyi, G. M., Kapetanovic, S., … Lockhart, P. J. (2019). Bioinformatics-based identification of expanded repeats: A non-reference intronic pentamer expansion in RFC1 causes CANVAS. *Am. J. Hum. Genet.*, *105*(1), 151–165. https://doi.org/10.1016/j.ajhg.2019.05.016

Robinson, J. T., Thorvaldsdottir, H., & Mesirov, J. P. (2026). Igv-reports: Embedding interactive genomic visualizations in HTML reports to aid variant review. *Bioinformatics*, *42*(4), btag125. https://doi.org/10.1093/bioinformatics/btag125

Schubach, M., Maass, T., Nazaretyan, L., Röner, S., & Kircher, M. (2024). CADD v1.7: Using protein language models, regulatory CNNs and other nucleotide-level scores to improve genome-wide variant predictions. *Nucleic Acids Res.*, *52*(D1), D1143–D1154. https://doi.org/10.1093/nar/gkad989

Stranneheim, H., Lagerstedt-Robinson, K., Magnusson, M., Kvarnung, M., Nilsson, D., Lesko, N., Engvall, M., Anderlid, B.-M., Arnell, H., Johansson, C. B., Barbaro, M., Björck, E., Bruhn, H., Eisfeldt, J., Freyer, C., Grigelioniene, G., Gustavsson, P., Hammarsjö, A., Hellström-Pigg, M., … Wedell, A. (2021). Integration of whole genome sequencing into a healthcare setting: High diagnostic rates across multiple clinical entities in 3219 rare disease patients. *Genome Med.*, *13*(1), 40. https://doi.org/10.1186/s13073-021-00855-5

Wallis, M., Bodek, S. D., Munro, J., Rafehi, H., Bennett, M. F., Ye, Z., Schneider, A., Gardiner, F., Valente, G., Murdoch, E., Uebergang, E., Hunter, J., Stutterd, C., Huq, A., Salmon, L., Scheffer, I., Eratne, D., Meyn, S., Fong, C. Y., … Berkovic, S. F. (2024). Experience of the first adult-focussed undiagnosed disease program in australia (AHA-UDP): Solving rare and puzzling genetic disorders is ageless. *Orphanet J. Rare Dis.*, *19*(1), 288. https://doi.org/10.1186/s13023-024-03297-5

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